Assignment No. 09: Logic Synthesis using Synopsys Design Compiler

FAROOQ ABDULLA
Online Internship on Analog and Digital VLSI Design Flow
Batch 01

Q1. Design and synthesize a 4-bit Universal Shift Register using Verilog HDL.

a) Implement all functional modes of operation.

b) Apply appropriate design constraints to achieve a synthesizable implementation.

c) Verify the functional correctness of the synthesized design.

d) Interpret the generated synthesis results with respect to area, timing, and power.

Q2. Design and synthesize an 8-bit Arithmetic Logic Unit (ALU) supporting arithmetic and logical operations.

a) Implement the complete functional specification of the ALU.

b) Analyze the synthesized implementation using the generated design reports.

c) Examine the technology mapped netlist and identify the major standard cells utilized.
d) Evaluate the quality of the synthesized design based on the obtained implementation results.

Q3. Design and synthesize a Sequence Detector for detecting the binary sequence 10110.

a) Develop an appropriate finite state machine for the required sequence.

b) Perform logic synthesis and validate the synthesized implementation.

c) Examine the technology mapped schematic and synthesized design hierarchy.

d) Analyze the generated synthesis reports and discuss the implementation characteristics of the synthesized design.

Q.1)

DESIGN CODE:

```
module usr_4bit (
    input wire clk,
    input wire rst_n,
    input wire [1:0] select,
    input wire shift_left_in,
    input wire shift_right_in,
    input wire [3:0] d,
    output reg [3:0] q       );

    always @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            q <= 4'b0000;
        end else begin
            case (select)
                2'b00: q <= q;
                2'b01: q <= {shift_right_in, q[3:1]};
                2'b10: q <= {q[2:0], shift_left_in};
                2'b11: q <= d;
                default: q <= q;
            endcase
        end
    end

endmodule
```

TESHBENCH CODE:

```
`timescale 1ns/1ps

module tb_usr_4bit;

    reg clk;
    reg rst_n;
    reg [1:0] select;
    reg shift_left_in;
    reg shift_right_in;
    reg [3:0] d;
    wire [3:0] q;

    usr_4bit uut (
        .clk(clk),
        .rst_n(rst_n),
        .select(select),
        .shift_left_in(shift_left_in),
```

```
        .shift_right_in(shift_right_in),
        .d(d),
        .q(q));

    always #10 clk = ~clk;

    initial begin

        $dumpfile("usr_simulation.vcd");
        $dumpvars(0, tb_usr_4bit);

        clk = 0;
        rst_n = 0;
        select = 2'b00;
        shift_left_in = 0;
        shift_right_in = 0;
        d = 4'b0000;

        #15 rst_n = 1;

        #10;
        select = 2'b11;
        d = 4'b1010;

        #20;
        select = 2'b00;
        d = 4'b1111;

        #20;
        select = 2'b01;
        shift_right_in = 1;
        #20;
        shift_right_in = 0;
        #40;

        #20;
        select = 2'b10;
        shift_left_in = 1;
        #20;
        shift_left_in = 0;
        #40;
        $display("Simulation complete. Waveform file generated.");
        $finish;
    end
endmodule
```

VCD FILE :

```
Compiler version W-2024.09; Runtime version W-2024.09;  Jul 17 20:44 2026
[SCL] SCL trace is enabled.
Checkout succeeded: VCS-BASE-RUNTIME/AF3AD4D7787CF4828637
        License file: 27020@14.139.1.126
        License Server: 27020@14.139.1.126
[SCL] 07/17/2026 20:44:34 Checking status for feature VCS-BASE-RUNTIME
[SCL] 07/17/2026 20:44:34 PID:210911 Client:smartlab6.calicut.nielit.in Server:27020@14.139.1.126 Checkout succ
RUNTIME 2024.09 from VCS-Base-Runtime-Pkg
         0 | 0 | 00 | 0 | 0 | 0000 | 0000
        15 | 1 | 00 | 0 | 0 | 0000 | 0000
        25 | 1 | 11 | 0 | 0 | 1010 | 0000
        30 | 1 | 11 | 0 | 0 | 1010 | 1010
        45 | 1 | 00 | 0 | 0 | 1111 | 1010
        65 | 1 | 01 | 0 | 1 | 1111 | 1010
        70 | 1 | 01 | 0 | 1 | 1111 | 1101
        85 | 1 | 01 | 0 | 0 | 1111 | 1101
        90 | 1 | 01 | 0 | 0 | 1111 | 0110
       110 | 1 | 01 | 0 | 0 | 1111 | 0011
       130 | 1 | 01 | 0 | 0 | 1111 | 0001
       145 | 1 | 10 | 1 | 0 | 1111 | 0001
       150 | 1 | 10 | 1 | 0 | 1111 | 0011
       165 | 1 | 10 | 0 | 0 | 1111 | 0011
       170 | 1 | 10 | 0 | 0 | 1111 | 0110
       190 | 1 | 10 | 0 | 0 | 1111 | 1100
Simulation complete.Waveform file generated
$finish called from file "../RTL/TB/tb_usr_4bit.v", line 66.
$finish at simulation time               205000
           V C S   S i m u l a t i o n   R e p o r t
Time: 205000 ps
CPU Time:      0.250 seconds;       Data structure size:   0.0Mb
Fri Jul 17 20:44:34 2026
```

REPORT:

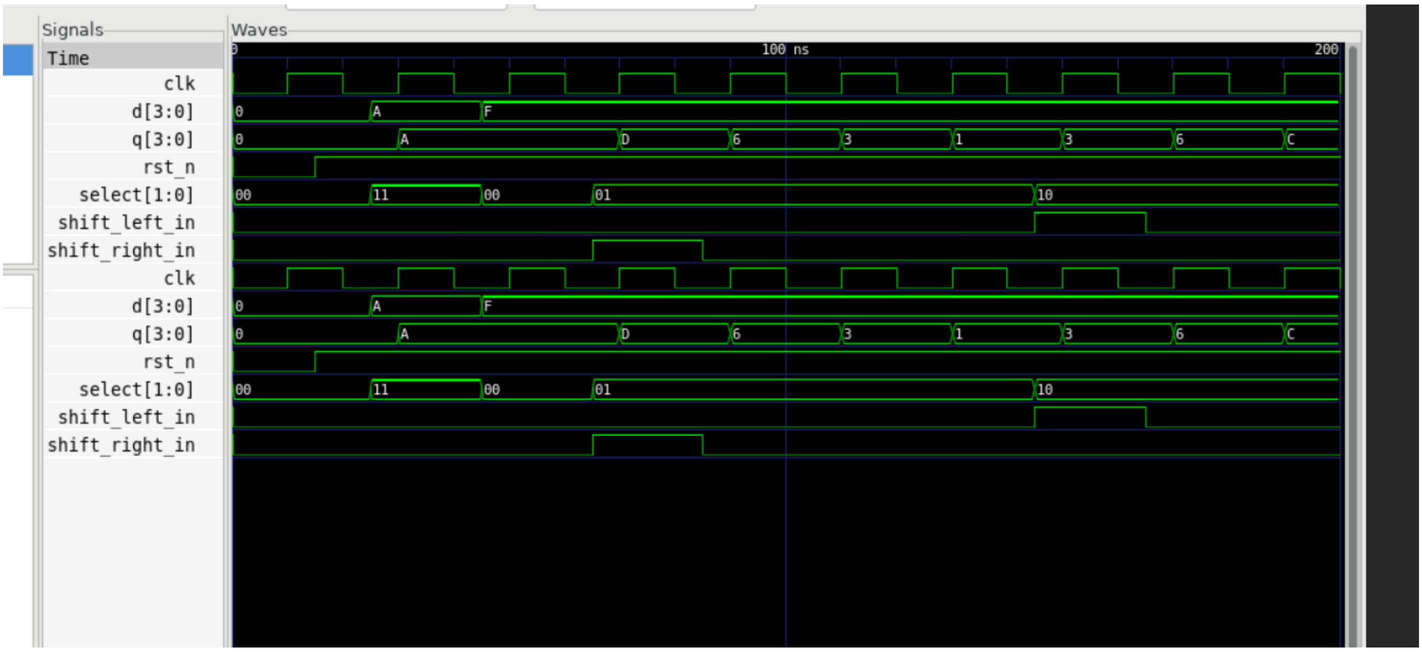
```
****************************************
Report : area
Design : seq_detector
Version: W-2024.09
Date   : Thu Jul 23 20:10:00 2026
****************************************

Library(s) Used:

    saed32rvt_ss0p95v125c (File: /home/int-05/Desktop/counter/tech/DB/saed32rvt_ss0p95v125c.db)

Number of ports:                            4
Number of nets:                            18
Number of cells:                           13
Number of combinational cells:              9
Number of sequential cells:                 3
Number of macros/black boxes:               0
Number of buf/inv:                          1
Number of references:                      10

Combinational area:                 20.331520
Buf/Inv area:                        1.270720
Noncombinational area:              21.348096
Macro/Black Box area:                0.000000
Net Interconnect area:               3.584609

Total cell area:                    41.679616
Total area:                         45.264226
1

****************************************
Report : power
        -analysis_effort low
Design : seq_detector
Version: W-2024.09
Date   : Thu Jul 23 20:10:00 2026
```

```
****************************************


Library(s) Used:

    saed32rvt_ss0p95v125c (File: /home/int-05/Desktop/counter/tech/DB/saed32rvt_ss0p95v125c.db)


Operating Conditions: ss0p95v125c   Library: saed32rvt_ss0p95v125c
Wire Load Model Mode: enclosed

Design        Wire Load Model            Library
------------------------------------------------
seq_detector           ForQA             saed32rvt_ss0p95v125c


Global Operating Voltage = 0.95
Power-specific unit information :
    Voltage Units = 1V
    Capacitance Units = 1.000000ff
    Time Units = 1ns
    Dynamic Power Units = 1uW    (derived from V,C,T units)
    Leakage Power Units = 1pW


Attributes
----------
i - Including register clock pin internal power


  Cell Internal Power  =  18.9164 uW   (98%)
  Net Switching Power  = 416.4846 nW    (2%)
                         ---------
Total Dynamic Power    =  19.3328 uW  (100%)

Cell Leakage Power     =   2.1057 uW


                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000            0.0000            0.0000            0.0000  (   0.00%)
memory             0.0000            0.0000            0.0000            0.0000  (   0.00%)
black_box          0.0000            0.0000            0.0000            0.0000  (   0.00%)
clock_network     16.5013            0.0000            0.0000           16.5013  (  76.97%)  i
register           1.9539            0.3077        1.4317e+06            3.6933  (  17.23%)
sequential         0.0000            0.0000            0.0000            0.0000  (   0.00%)
combinational      0.4612            0.1088        6.7407e+05            1.2440  (   5.80%)
--------------------------------------------------------------------------------------------------
Total             18.9164 uW         0.4165 uW     2.1057e+06 pW        21.4386 uW
1

nformation: Updating design information... (UID-85)

****************************************
Report : qor
Design : seq_detector
Version: W-2024.09
Date   : Thu Jul 23 20:10:00 2026
****************************************


  Timing Path Group 'clk'
  -----------------------------------
  Levels of Logic:              3.00
  Critical Path Length:         0.38
  Critical Path Slack:          0.56
  Critical Path Clk Period:     1.00
  Total Negative Slack:         0.00
  No. of Violating Paths:       0.00
```

```
  Worst Hold Violation:          0.00
  Total Hold Violation:          0.00
  No. of Hold Violations:        0.00
  -----------------------------------


  Cell Count
  -----------------------------------
  Hierarchical Cell Count:          0
  Hierarchical Port Count:          0
  Leaf Cell Count:                 12
  Buf/Inv Cell Count:               1
  Buf Cell Count:                   0
  Inv Cell Count:                   1
  CT Buf/Inv Cell Count:            0
  Combinational Cell Count:         9
  Sequential Cell Count:            3
  Macro Count:                      0
  -----------------------------------


  Area
  -----------------------------------
  Combinational Area:       20.331520
  Noncombinational Area:    21.348096
  Buf/Inv Area:              1.270720
  Total Buffer Area:             0.00
  Total Inverter Area:           1.27
  Macro/Black Box Area:      0.000000
  Net Area:                  3.584609
  -----------------------------------
  Cell Area:                41.679616
  Design Area:              45.264226


  Design Rules
  -----------------------------------
  Total Number of Nets:            18
  Nets With Violations:             0
  Max Trans Violations:             0
  Max Cap Violations:               0
  -----------------------------------


  Hostname: smartlab6.calicut.nielit.in

  Compile CPU Statistics
  -----------------------------------------
  Resource Sharing:                    0.01
  Logic Optimization:                  0.03
  Mapping Optimization:                0.17
  -----------------------------------------
  Overall Compile Time:                2.76
  Overall Compile Wall Clock Time:     3.02

  --------------------------------------------------------------------

  Design  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0


  Design (Hold)  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0

  --------------------------------------------------------------------


1

****************************************
Report : timing
        -path full
```

```
        -delay max
        -max_paths 1
Design : seq_detector
Version: W-2024.09
Date   : Thu Jul 23 20:10:00 2026
****************************************

Operating Conditions: ss0p95v125c   Library: saed32rvt_ss0p95v125c
Wire Load Model Mode: enclosed

  Startpoint: current_state_reg[0]
              (rising edge-triggered flip-flop clocked by clk)
  Endpoint: current_state_reg[1]
            (rising edge-triggered flip-flop clocked by clk)
  Path Group: clk
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  seq_detector       ForQA                 saed32rvt_ss0p95v125c

  Point                                                   Incr       Path
  --------------------------------------------------------------------------
  clock clk (rise edge)                                   0.00       0.00
  clock network delay (ideal)                             0.00       0.00
  current_state_reg[0]/CLK (DFFARX1_RVT)                  0.00       0.00 r
  current_state_reg[0]/Q (DFFARX1_RVT)                    0.18       0.18 r
  U13/Y (NOR4X1_RVT)                                      0.13       0.31 f
  U18/Y (INVX1_RVT)                                       0.02       0.34 r
  U21/Y (NAND3X0_RVT)                                     0.04       0.38 f
  current_state_reg[1]/D (DFFARX1_RVT)                    0.00       0.38 f
  data arrival time                                                  0.38

  clock clk (rise edge)                                   1.00       1.00
  clock network delay (ideal)                             0.00       1.00
  current_state_reg[1]/CLK (DFFARX1_RVT)                  0.00       1.00 r
  library setup time                                     -0.07       0.93
  data required time                                                 0.93
  --------------------------------------------------------------------------
  data required time                                                 0.93
  data arrival time                                                 -0.38
  --------------------------------------------------------------------------
  slack (MET)                                                        0.56


1
```

GTKWave:



2.Q)

DESIGN CODE:

```
`timescale 1ns/1ps
module alu_8bit (
    input  wire [7:0] a,
    input  wire [7:0] b,
    input  wire [2:0] opcode,
    output reg  [7:0] alu_out,
    output reg        carry_out  );

    always @(*) begin

        alu_out   = 8'b0000_0000;
        carry_out = 1'b0;

        case (opcode)
            3'b000: begin
                {carry_out, alu_out} = a + b;
            end
            3'b001: begin
                alu_out = a - b;
                carry_out = (a < b);
            end
            3'b010: begin
                alu_out = a & b;
            end
```

```
            3'b011: begin
                alu_out = a | b;
            end
            3'b100: begin
                alu_out = a ^ b;
            end
            3'b101: begin
                alu_out = a << 1;
            end
            3'b110: begin
                alu_out = a >> 1;
            end
            3'b111: begin
                alu_out = a;
            end
            default: begin
                alu_out = a;
            end
        endcase
    end
endmodule
```

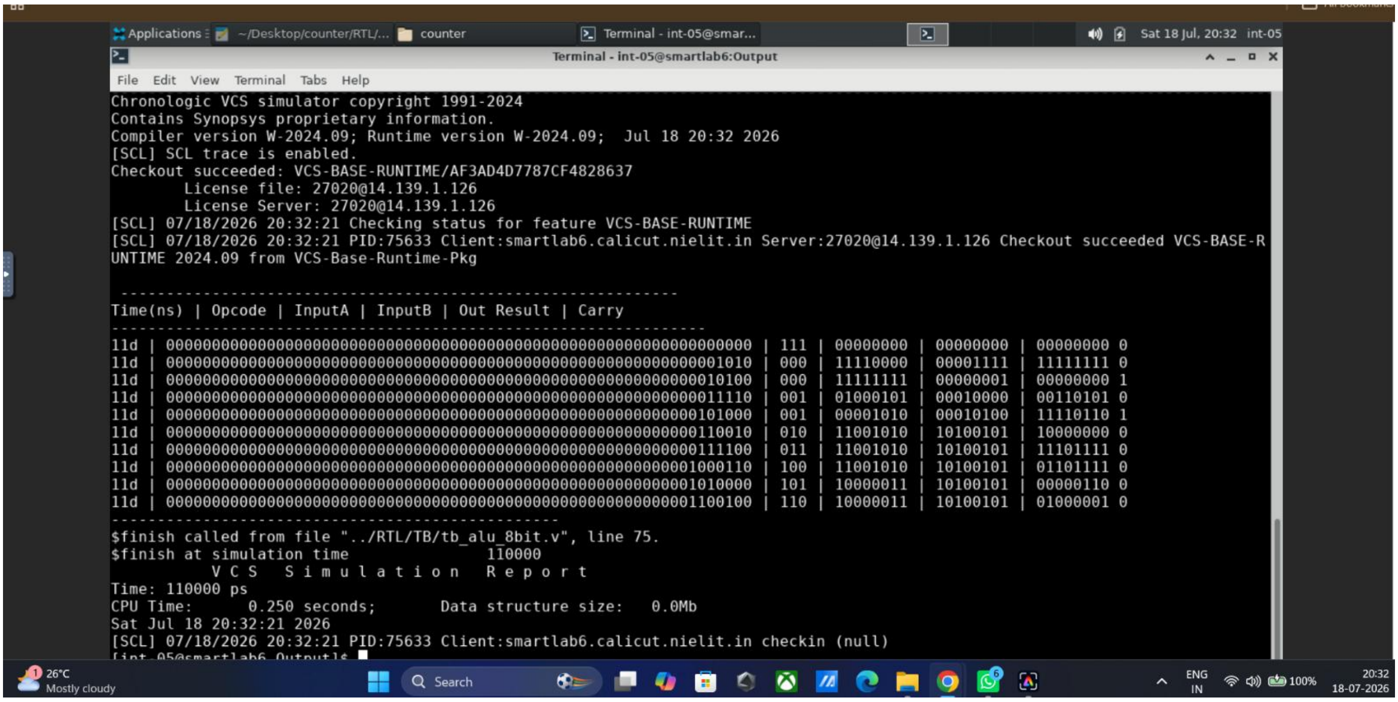
TESTBENCH CODE:

```
`timescale 1ns/1ps

module tb_alu_8bit;

    reg [7:0] a;
    reg [7:0] b;
    reg [2:0] opcode;
    wire [7:0] alu_out;
    wire carry_out;

    alu_8bit uut (
        .a(a),
        .b(b),
        .opcode(opcode),
        .alu_out(alu_out),
        .carry_out(carry_out) );


    initial begin
        $display("\n-----------------------------------------------------------");
        $display("   Time(ns) | Opcode |     Input A    |     Input B    | Out
Result | Carry");
        $display("-------------------------------------------------------------");
```

```
        $monitor("%11d |    %b   |    %b    |    %b    |  %b  |    %b",
                 $time, opcode, a, b, alu_out, carry_out);
    end

    initial begin

        $dumpfile("alu_simulation.vcd");
        $dumpvars(0, tb_alu_8bit);

        a = 8'h00; b = 8'h00; opcode = 3'b111;

        #10;
        opcode = 3'b000;
        a = 8'hF0; b = 8'h0F;
        #10;
        a = 8'hFF; b = 8'h01;

        #10;
        opcode = 3'b001;
        a = 8'h45; b = 8'h10;
        #10;
        a = 10;    b = 20;

        #10;
        opcode = 3'b010;
        a = 8'b1100_1010; b = 8'b1010_0101;

        #10;
        opcode = 3'b011;
        a = 8'b1100_1010; b = 8'b1010_0101;

        #10;
        opcode = 3'b100;
        a = 8'b1100_1010; b = 8'b1010_0101;

        #10;
        opcode = 3'b101;
        a = 8'b1000_0011;

        #10;
        opcode = 3'b110;
        a = 8'b1000_0011;

        #10;
        $display("------------------------------------------------------------");
        $finish;
```

```
    end
endmodule
```

VCD FILE :



```
Chronologic VCS simulator copyright 1991-2024
Contains Synopsys proprietary information.
Compiler version W-2024.09; Runtime version W-2024.09;  Jul 18 20:32 2026
[SCL] SCL trace is enabled.
Checkout succeeded: VCS-BASE-RUNTIME/AF3AD4D7787CF4828637
        License file: 27020@14.139.1.126
        License Server: 27020@14.139.1.126
[SCL] 07/18/2026 20:32:21 Checking status for feature VCS-BASE-RUNTIME
[SCL] 07/18/2026 20:32:21 PID:75633 Client:smartlab6.calicut.nielit.in Server:27020@14.139.1.126 Checkout succeeded VCS-BASE-R
UNTIME 2024.09 from VCS-Base-Runtime-Pkg

 ------------------------------------------------------------
Time(ns) | Opcode | InputA | InputB | Out Result | Carry
-----------------------------------------------------------------
11d | 0000000000000000000000000000000000000000000000000000000000000000 | 111 | 00000000 | 00000000 | 00000000 0
11d | 0000000000000000000000000000000000000000000000000000000000001010 | 000 | 11110000 | 00001111 | 11111111 0
11d | 0000000000000000000000000000000000000000000000000000000000010100 | 000 | 11111111 | 00000001 | 00000000 1
11d | 0000000000000000000000000000000000000000000000000000000000011110 | 001 | 01000101 | 00010000 | 00110101 0
11d | 0000000000000000000000000000000000000000000000000000000000101000 | 001 | 00001010 | 00010100 | 11110110 1
11d | 0000000000000000000000000000000000000000000000000000000000110010 | 010 | 11001010 | 10100101 | 10000000 0
11d | 0000000000000000000000000000000000000000000000000000000000111100 | 011 | 11001010 | 10100101 | 11101111 0
11d | 0000000000000000000000000000000000000000000000000000000001000110 | 100 | 11001010 | 10100101 | 01101111 0
11d | 0000000000000000000000000000000000000000000000000000000001010000 | 101 | 10000011 | 10100101 | 00000110 0
11d | 0000000000000000000000000000000000000000000000000000000001100100 | 110 | 10000011 | 10100101 | 01000001 0
-----------------------------------------------
$finish called from file "../RTL/TB/tb_alu_8bit.v", line 75.
$finish at simulation time               110000
           V C S   S i m u l a t i o n   R e p o r t
Time: 110000 ps
CPU Time:      0.250 seconds;       Data structure size:   0.0Mb
Sat Jul 18 20:32:21 2026
[SCL] 07/18/2026 20:32:21 PID:75633 Client:smartlab6.calicut.nielit.in checkin (null)
```

REPORT:

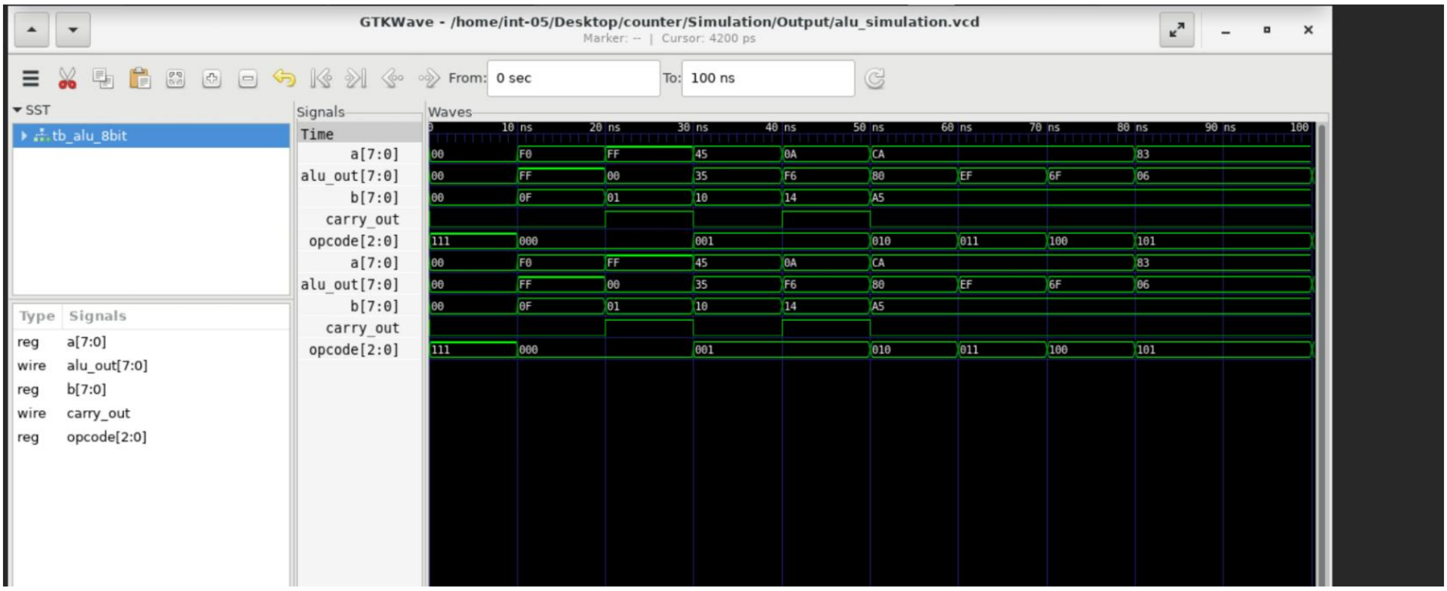
```
****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : alu_8bit
Version: W-2024.09
Date   : Thu Jul 23 20:02:29 2026
****************************************

Operating Conditions: ss0p95v125c   Library: saed32rvt_ss0p95v125c
Wire Load Model Mode: enclosed

  Startpoint: a[0] (input port)
  Endpoint: alu_out[7] (output port)
  Path Group: (none)
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  alu_8bit           8000                  saed32rvt_ss0p95v125c

  Point                                    Incr       Path
  -----------------------------------------------------------
  input external delay                     0.00       0.00 f
  a[0] (in)                                0.00       0.00 f
  U111/Y (INVX0_RVT)                       0.01       0.01 r
  U112/Y (AND2X1_RVT)                      0.06       0.07 r
  U114/Y (AO222X1_RVT)                     0.15       0.22 r
  U116/Y (AO222X1_RVT)                     0.16       0.37 r
  U118/Y (AO222X1_RVT)                     0.16       0.53 r
  U120/Y (AO222X1_RVT)                     0.16       0.68 r
  U122/Y (AO222X1_RVT)                     0.16       0.84 r
  U123/Y (AO222X1_RVT)                     0.13       0.97 r
```

```
  U191/Y (AO21X1_RVT)                       0.08       1.05 r
  U192/Y (AOI221X1_RVT)                     0.14       1.19 f
  U193/Y (OA22X1_RVT)                       0.07       1.26 f
  U195/Y (NAND3X0_RVT)                      0.03       1.29 r
  alu_out[7] (out)                          0.00       1.29 r
  data arrival time                                    1.29
  -----------------------------------------------------------
  (Path is unconstrained)


1
****************************************
Report : power
        -analysis_effort low
Design : alu_8bit
Version: W-2024.09
Date   : Thu Jul 23 20:02:29 2026
****************************************


Library(s) Used:

    saed32rvt_ss0p95v125c (File: /home/int-05/Desktop/counter/tech/DB/saed32rvt_ss0p95v125c.db)


Operating Conditions: ss0p95v125c   Library: saed32rvt_ss0p95v125c
Wire Load Model Mode: enclosed

Design        Wire Load Model            Library
------------------------------------------------
alu_8bit               8000              saed32rvt_ss0p95v125c


Global Operating Voltage = 0.95
Power-specific unit information :
    Voltage Units = 1V
    Capacitance Units = 1.000000ff
    Time Units = 1ns
    Dynamic Power Units = 1uW    (derived from V,C,T units)
    Leakage Power Units = 1pW


Attributes
----------
i - Including register clock pin internal power


  Cell Internal Power  =  17.6987 uW   (68%)
  Net Switching Power  =   8.3499 uW   (32%)
                         ---------
Total Dynamic Power    =  26.0486 uW  (100%)

Cell Leakage Power     =   9.6924 uW

Information: report_power power group summary does not include estimated clock tree power. (PWR-789)

                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000            0.0000            0.0000  (   0.00%)
memory             0.0000           0.0000            0.0000            0.0000  (   0.00%)
black_box          0.0000           0.0000            0.0000            0.0000  (   0.00%)
clock_network      0.0000           0.0000            0.0000            0.0000  (   0.00%)  i
register           0.0000           0.0000            0.0000            0.0000  (   0.00%)
sequential         0.0000           0.0000            0.0000            0.0000  (   0.00%)
combinational     17.6987           8.3499        9.6924e+06           35.7411  ( 100.00%)
--------------------------------------------------------------------------------------------------
Total             17.6987 uW        8.3499 uW     9.6924e+06 pW        35.7411 uW
1
```

```
Information: Updating design information... (UID-85)

****************************************
Report : qor
Design : alu_8bit
Version: W-2024.09
Date   : Thu Jul 23 20:02:29 2026
****************************************


  Timing Path Group (none)
  -----------------------------------
  Levels of Logic:              12.00
  Critical Path Length:          1.29
  Critical Path Slack:         uninit
  Critical Path Clk Period:       n/a
  Total Negative Slack:          0.00
  No. of Violating Paths:        0.00
  Worst Hold Violation:          0.00
  Total Hold Violation:          0.00
  No. of Hold Violations:        0.00
  -----------------------------------

  Cell Count
  -----------------------------------
  Hierarchical Cell Count:          0
  Hierarchical Port Count:          0
  Leaf Cell Count:                130
  Buf/Inv Cell Count:              28
  Buf Cell Count:                   0
  Inv Cell Count:                  28
  CT Buf/Inv Cell Count:            0
  Combinational Cell Count:       130
  Sequential Cell Count:            0
  Macro Count:                      0
  -----------------------------------

  Area
  -----------------------------------
  Combinational Area:      285.403713
  Noncombinational Area:     0.000000
  Buf/Inv Area:             35.580160
  Total Buffer Area:             0.00
  Total Inverter Area:          35.58
  Macro/Black Box Area:      0.000000
  Net Area:                 78.495911
  -----------------------------------
  Cell Area:               285.403713
  Design Area:             363.899624


  Design Rules
  -----------------------------------
  Total Number of Nets:           150
  Nets With Violations:             0
  Max Trans Violations:             0
  Max Cap Violations:               0
  -----------------------------------


  Hostname: smartlab6.calicut.nielit.in

  Compile CPU Statistics
  -----------------------------------------
  Resource Sharing:                    0.06
  Logic Optimization:                  0.03
  Mapping Optimization:                0.21
  -----------------------------------------
  Overall Compile Time:                2.18
  Overall Compile Wall Clock Time:     2.45
```

```
  -------------------------------------------------------------------

  Design  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0


  Design (Hold)  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0

  -------------------------------------------------------------------

****************************************
Report : area
Design : alu_8bit
Version: W-2024.09
Date   : Thu Jul 23 20:02:29 2026
****************************************

Library(s) Used:

    saed32rvt_ss0p95v125c (File: /home/int-05/Desktop/counter/tech/DB/saed32rvt_ss0p95v125c.db)

Number of ports:                           28
Number of nets:                           150
Number of cells:                          131
Number of combinational cells:            130
Number of sequential cells:                 0
Number of macros/black boxes:               0
Number of buf/inv:                         28
Number of references:                      17

Combinational area:                285.403713
Buf/Inv area:                       35.580160
Noncombinational area:               0.000000
Macro/Black Box area:                0.000000
Net Interconnect area:              78.495911

Total cell area:                   285.403713
Total area:                        363.899624
1
```

GTKWave:



Q.3)

Q.1)

DESIGN CODE:

```
module seq_detector (
    input wire clk,
    input wire rst_n,
    input wire din,
    output reg dout
);

    // State Declarations using One-Hot-like Encoding parameters
    localparam [2:0] S_IDLE = 3'b000,
                     S_1    = 3'b001,
                     S_10   = 3'b010,
                     S_101  = 3'b011,
                     S_1011 = 3'b100;

    reg [2:0] current_state, next_state;

    // 1. Sequential State Register Block
    always @(posedge clk or negedge rst_n) begin
        if (!rst_n)
            current_state <= S_IDLE;
        else
            current_state <= next_state;
    end
```

```
    // 2. Next-State Combinational Combinational Logic Block
    always @(*) begin
        case (current_state)
            S_IDLE: next_state = (din) ? S_1    : S_IDLE;
            S_1   : next_state = (din) ? S_1    : S_10;
            S_10  : next_state = (din) ? S_101  : S_IDLE;
            S_101 : next_state = (din) ? S_1011 : S_10;
            S_1011: next_state = (din) ? S_1    : S_10;
            default: next_state = S_IDLE;
        endcase
    end

    // 3. Mealy Output Combinational Logic Block
    always @(*) begin
        if ((current_state == S_1011) && (din == 1'b0))
            dout = 1'b1;
        else
            dout = 1'b0;
    end

endmodule
```

TESHBENCH CODE:

```
`timescale 1ns/1ps

module tb_seq_detector;
    reg clk;
    reg rst_n;
    reg din;
    wire dout;

    seq_detector uut (
        .clk(clk),
        .rst_n(rst_n),
        .din(din),
        .dout(dout)
    );

    always #5 clk = ~clk; // 100MHz Clock

    initial begin
        $dumpfile("seq_sim.vcd");
        $dumpvars(0, tb_seq_detector);

        $monitor("Time=%4d ns | Reset=%b | Din=%b | Dout=%b | State=%b",
                 $time, rst_n, din, dout, uut.current_state);
```

```
        clk = 0; rst_n = 0; din = 0;
        #12 rst_n = 1;

        // Feed Sequence: 1 -> 0 -> 1 -> 1 -> 0 (Detection 1)
        @(posedge clk); din = 1;
        @(posedge clk); din = 0;
        @(posedge clk); din = 1;
        @(posedge clk); din = 1;
        @(posedge clk); din = 0; // dout should hit 1 here

        // Test Overlapping: Feed remaining bits to form another 10110
        // Current sequence tail is ..10. Add 1 -> 1 -> 0
        @(posedge clk); din = 1;
        @(posedge clk); din = 1;
        @(posedge clk); din = 0; // dout hits 1 again (overlapping verified)

        #20;
        $finish;
    end
endmodule
```

REPORT:

```
****************************************
Report : area
Design : seq_detector
Version: W-2024.09
Date   : Thu Jul 23 20:10:00 2026
****************************************

Library(s) Used:

    saed32rvt_ss0p95v125c (File: /home/int-05/Desktop/counter/tech/DB/saed32rvt_ss0p95v125c.db)

Number of ports:                            4
Number of nets:                            18
Number of cells:                           13
Number of combinational cells:              9
Number of sequential cells:                 3
Number of macros/black boxes:               0
Number of buf/inv:                          1
Number of references:                      10

Combinational area:                 20.331520
Buf/Inv area:                        1.270720
Noncombinational area:              21.348096
Macro/Black Box area:                0.000000
Net Interconnect area:               3.584609

Total cell area:                    41.679616
Total area:                         45.264226
1

****************************************
Report : power
```

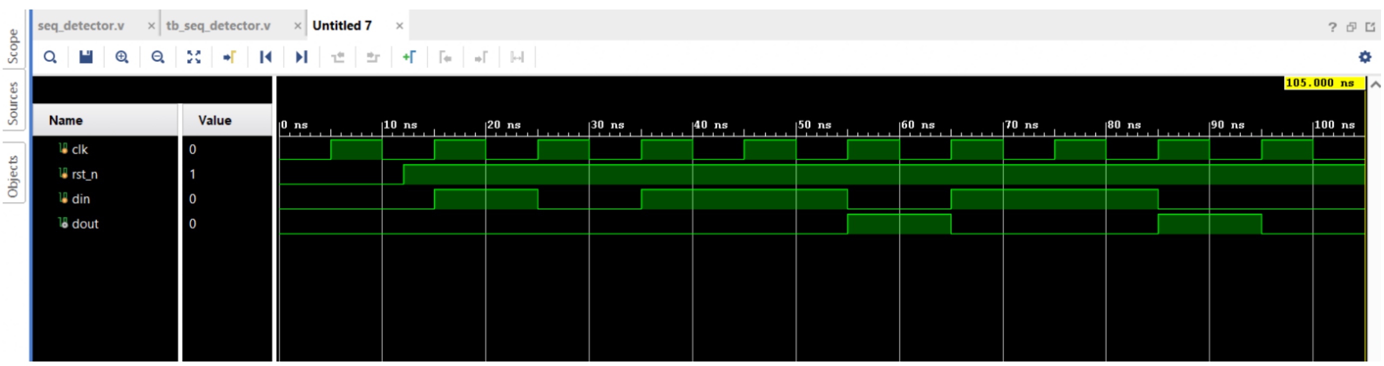
```
        -analysis_effort low
Design : seq_detector
Version: W-2024.09
Date   : Thu Jul 23 20:10:00 2026
****************************************


Library(s) Used:

    saed32rvt_ss0p95v125c (File: /home/int-05/Desktop/counter/tech/DB/saed32rvt_ss0p95v125c.db)


Operating Conditions: ss0p95v125c   Library: saed32rvt_ss0p95v125c
Wire Load Model Mode: enclosed

Design        Wire Load Model            Library
------------------------------------------------
seq_detector           ForQA             saed32rvt_ss0p95v125c


Global Operating Voltage = 0.95
Power-specific unit information :
    Voltage Units = 1V
    Capacitance Units = 1.000000ff
    Time Units = 1ns
    Dynamic Power Units = 1uW    (derived from V,C,T units)
    Leakage Power Units = 1pW


Attributes
----------
i - Including register clock pin internal power


  Cell Internal Power  =  18.9164 uW   (98%)
  Net Switching Power  = 416.4846 nW    (2%)
                         ---------
Total Dynamic Power    =  19.3328 uW  (100%)

Cell Leakage Power     =   2.1057 uW


                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000            0.0000            0.0000            0.0000  (   0.00%)
memory             0.0000            0.0000            0.0000            0.0000  (   0.00%)
black_box          0.0000            0.0000            0.0000            0.0000  (   0.00%)
clock_network     16.5013            0.0000            0.0000           16.5013  (  76.97%)  i
register           1.9539            0.3077        1.4317e+06            3.6933  (  17.23%)
sequential         0.0000            0.0000            0.0000            0.0000  (   0.00%)
combinational      0.4612            0.1088        6.7407e+05            1.2440  (   5.80%)
--------------------------------------------------------------------------------------------------
Total             18.9164 uW         0.4165 uW     2.1057e+06 pW        21.4386 uW
1

nformation: Updating design information... (UID-85)

****************************************
Report : qor
Design : seq_detector
Version: W-2024.09
Date   : Thu Jul 23 20:10:00 2026
****************************************


  Timing Path Group 'clk'
  -----------------------------------
  Levels of Logic:               3.00
  Critical Path Length:          0.38
```

```
Critical Path Slack:           0.56
Critical Path Clk Period:      1.00
Total Negative Slack:          0.00
No. of Violating Paths:        0.00
Worst Hold Violation:          0.00
Total Hold Violation:          0.00
No. of Hold Violations:        0.00
-----------------------------------


Cell Count
-----------------------------------
Hierarchical Cell Count:          0
Hierarchical Port Count:          0
Leaf Cell Count:                 12
Buf/Inv Cell Count:               1
Buf Cell Count:                   0
Inv Cell Count:                   1
CT Buf/Inv Cell Count:            0
Combinational Cell Count:         9
Sequential Cell Count:            3
Macro Count:                      0
-----------------------------------


Area
-----------------------------------
Combinational Area:       20.331520
Noncombinational Area:    21.348096
Buf/Inv Area:              1.270720
Total Buffer Area:             0.00
Total Inverter Area:           1.27
Macro/Black Box Area:      0.000000
Net Area:                  3.584609
-----------------------------------
Cell Area:                41.679616
Design Area:              45.264226


Design Rules
-----------------------------------
Total Number of Nets:            18
Nets With Violations:             0
Max Trans Violations:             0
Max Cap Violations:               0
-----------------------------------


Hostname: smartlab6.calicut.nielit.in

Compile CPU Statistics
-----------------------------------------
Resource Sharing:                    0.01
Logic Optimization:                  0.03
Mapping Optimization:                0.17
-----------------------------------------
Overall Compile Time:                2.76
Overall Compile Wall Clock Time:     3.02

--------------------------------------------------------------------

Design  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0


Design (Hold)  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0

--------------------------------------------------------------------
```

1

```
****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : seq_detector
Version: W-2024.09
Date   : Thu Jul 23 20:10:00 2026
****************************************

Operating Conditions: ss0p95v125c   Library: saed32rvt_ss0p95v125c
Wire Load Model Mode: enclosed

  Startpoint: current_state_reg[0]
              (rising edge-triggered flip-flop clocked by clk)
  Endpoint: current_state_reg[1]
            (rising edge-triggered flip-flop clocked by clk)
  Path Group: clk
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  seq_detector       ForQA                 saed32rvt_ss0p95v125c

  Point                                                   Incr       Path
  --------------------------------------------------------------------------
  clock clk (rise edge)                                   0.00       0.00
  clock network delay (ideal)                             0.00       0.00
  current_state_reg[0]/CLK (DFFARX1_RVT)                  0.00       0.00 r
  current_state_reg[0]/Q (DFFARX1_RVT)                    0.18       0.18 r
  U13/Y (NOR4X1_RVT)                                      0.13       0.31 f
  U18/Y (INVX1_RVT)                                       0.02       0.34 r
  U21/Y (NAND3X0_RVT)                                     0.04       0.38 f
  current_state_reg[1]/D (DFFARX1_RVT)                    0.00       0.38 f
  data arrival time                                                  0.38

  clock clk (rise edge)                                   1.00       1.00
  clock network delay (ideal)                             0.00       1.00
  current_state_reg[1]/CLK (DFFARX1_RVT)                  0.00       1.00 r
  library setup time                                     -0.07       0.93
  data required time                                                 0.93
  --------------------------------------------------------------------------
  data required time                                                 0.93
  data arrival time                                                 -0.38
  --------------------------------------------------------------------------
  slack (MET)                                                        0.56


1
```

SIMULATION:



SCHEMATIC: